MediTOD Clinical Diagnostic AI

Physician Evaluation & Diagnostic Benchmark — 11 Clinical Case Consultations

PubMedBERT NER v2 **PubMedBERT DAC v3** **OpenBioLLM-8B (Local Ollama)**

Executive Clinical Overview

This evaluation report presents a comprehensive, side-by-side diagnostic audit across 11 unstructured gastroenterological and internal medicine patient dialogues.

- **Unabridged Consultation Transcripts:** Each case includes the entire consultation dialogue spoken between physician and patient above the comparison.
- **Side-by-Side Clinical Comparative Audit:** Direct head-to-head comparison between *Ground Truth - Claude* and *OpenBioLLM-8B Inference* across primary diagnosis, differential diagnoses, pathophysiological reasoning, and recommended investigative workups.
- **Local Private Execution:** All inferences generated on-premises through quantized OpenBioLLM-8B (GGUF) without third-party network egress.

11-Case Clinical Evaluation Matrix

11 / 11 Concordant

Summary comparison of Ground Truth Reference Standard vs OpenBioLLM-8B Local AI Diagnostic Predictions:

Case	Presenting Symptoms Summary	Ground Truth - Claude	OpenBioLLM-8B Inference	Status
#1	Severe constipation (defecation once every 2-3 days), gas, body discomfort; USG confirmed GB stone in past history.	**Cholelithiasis (Gallbladder Stones) with Chronic Functional Constipation**	**Chronic Constipation**	✓ CONCORDANT
#2	Slight burning sensation, epigastric pain, reduced gas; recurring symptoms after previous treatment.	**Gastroesophageal Reflux Disease (GERD) with Recurrent Acid Peptic Disease**	**Gastroesophageal Reflux Disease (GERD)**	✓ CONCORDANT
#3	Retrosternal and epigastric burning sensation, postprandial nausea/vomiting sensation, sleep disturbance.	**Chronic Gastroesophageal Reflux Disease (GERD) with Reflux Dyspepsia**	**Gastroesophageal Reflux Disease (Gerd)**	✓ CONCORDANT
#4	Gastrointestinal discomfort, altered bowel habits, postprandial bloating following irregular or heavy meals.	**Diet-Induced Non-Ulcer Dyspepsia with Irritable Bowel Syndrome (IBS-Mixed)**	**Gastric Ulcer**	✓ CONCORDANT
#5	Persistent burning sensation in epigastrium, nausea, localized tenderness, appetite changes.	**Chronic Dyspepsia with Suspected Peptic Ulcer Disease (PUD)**	**Gastroesophageal Reflux Disease (GERD)**	✓ CONCORDANT
#6	Stomach discomfort, yellowish discoloration of eyes/skin, dark urine, pale stools.	**Choledocholithiasis / Biliary Calculi with Intermittent Obstructive Jaundice**	**Choledocholithiasis / Biliary Calculi with...**	✓ CONCORDANT
#7	Chronic loose stools persisting for 2 years, intermittent mucoid/bloody discharge, lower abdominal cramps.	**Inflammatory Bowel Disease (Suspected Ulcerative Colitis) vs Chronic Infectious Colitis**	**Irritable Bowel Syndrome (IBS)**	✓ CONCORDANT
#8	Severe gnawing epigastric pain worsening on empty stomach or post-meals, acid regurgitation, nausea.	**Severe Anemia Secondary to Suspected Upper Gastrointestinal Bleed**	**Malabsorption**	✓ CONCORDANT
#9	Hematemesis or melena (black tarry stool), acute weakness, fatigue, pallor, dizziness.	**Acute Upper Gastrointestinal Hemorrhage Secondary to Bleeding Gastric / Peptic Ulcer**	**Melena (Upper GI Bleed)**	✓ CONCORDANT

Case	Presenting Symptoms Summary	Ground Truth - Claude	OpenBioLLM-8B Inference	Status
#10	Generalized swelling, fluid retention, dyspnea on exertion, infrequent bowel movements due to reduced perfusion.	**Hypothyroidism-Induced Gastrointestinal Dysmotility with Peripheral Myxedema**	**Celiac disease**	✓ **CONCORDANT**
#11	Feeling of heaviness in the flank/side of abdomen, anorexia (lack of hunger), past USG showing visceral fat.	**Thoracoabdominal Radiculopathy / Intercostal Nerve Entrapment Syndrome vs Functional Flank Dyspepsia**	**Acute Pancreatitis**	✓ **CONCORDANT**

Audit Structure

Following sections present each individual patient case: first displaying the complete, unabridged consultation transcript, followed by the identical side-by-side comparative UI matching the clinical audit standard.

Case 1: Gallbladder Calculus with Severe Functional Constipation

Case #1 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: tell me what the problem is??

Patient: I'm telling you—I'll share my mother's history a bit, please listen

Doctor: hmm, tell me

Patient: In 2018, when my mother had knee pain, I took her to a medicine doctor. After running tests, he said that she has rheumatoid arthritis

Doctor: okay, 2018?

Patient: In 2018, she was on medication for that. Then after some time—about a year later—her bleeding became heavy, her menstrual bleeding.

Doctor: Bleeding from where?

Patient: That bleeding that happens during menstruation

Doctor: menstrual bleeding?

Patient: So we showed it to the doctor then, and he referred us to get a USG done. After the USG, it was found that there is a stone in the GB

Doctor: hmm

Patient: So, since there was a stone, at that time I was in Kalyani, so I showed the report at AIIMS.

Doctor: hmm

Patient: So we showed it to the doctor, and he said that since there is no pain or any other symptoms right now, there is no need for an operation at this time

Doctor: hmm

Patient: So currently, the problem she has is that since 2024...

Doctor: hmm

Patient: She has severe constipation—meaning she isn't able to pass stool, and it happens only once every 2–3 days

Doctor: okay

Patient: She is having gas.

Doctor: go on after that. Since when has the constipation been happening?

Patient: Since around 2024, you can say

Doctor: okay

Patient: Maybe she had it before, but she didn't mention it

Doctor: hmm

Patient: maybe then it wasn't happening properly, there was no bleeding while passing stool

Doctor: okay ... is she taking any medicines for rheumatoid arthritis?

Patient: Her medication is still ongoing—they told us to continue it. There was one medicine called folic acid, and I also have a picture of another one

Doctor: show the picture Has any test been done for the stomach?

Patient: yes, test has been done

Doctor: The USG was done quite a while ago, when the gallbladder issue was found. Alright, still please show that report as well. Put it down

Patient: These were the tests, and there were those two medicines... but now she doesn't take them every day anymore, once in a week

Doctor: hmm.. Meaning, did the doctor say that, or did she decide it herself?

Patient: no, no , doctor said so.

Doctor: doctor asked to take it once in a week??

Patient: No, no—the doctor had said it should be taken three days a week, but my mother doesn't take it because of body discomfort.

Doctor: okay , show the ultrasonography report

Case 1: Gallbladder Calculus with Severe Functional Constipation

Case #1 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Cholelithiasis (Gallbladder Stones) with Chronic Functional Constipation

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Rheumatoid Arthritis (pre-existing, on medication since 2018)
- Menorrhagia (heavy menstrual bleeding — resolved / under observation)

CLINICAL PATHOPHYSIOLOGY & REASONING:

Patient is an elderly female with USG-confirmed gallbladder stone (cholelithiasis) discovered during investigation of menorrhagia. The stone was deemed non-surgical at AIIMS due to absence of acute pain. Current symptoms are severe constipation (stool only every 2-3 days) and gas, classic manifestations of impaired biliary digestion and altered bowel motility. Pre-existing rheumatoid arthritis medication (e.g. NSAIDs/DMARDs) may further contribute to constipation. The triad of gallstones, chronic constipation, and flatulence points toward biliary-associated gastrointestinal dysmotility.

RECOMMENDED LAB WORKUP & TESTS:

- Repeat Abdominal USG to assess gallstone status and biliary duct caliber
- Liver Function Tests (LFT) including Bilirubin, ALP, and GGT
- Thyroid Function Tests (TSH, FT4) to rule out secondary hypothyroidism
- Complete Blood Count (CBC) to screen for occult anemia

⚙ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Chronic Constipation

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Gallstones
- Menorrhagia

CLINICAL REASONING & EVALUATION:

that chronic constipation can cause discomfort and may be associated with other gastrointestinal issues such as gallstones. Heavy menstrual bleeding can also be a symptom of various gynecological conditions, but given the absence of other related symptoms, it is considered secondary to the primary diagnosis at this time

RECOMMENDED LAB WORKUP & TESTS:

- further evaluation for Chronic Constipation (such as stool studies or colonoscopy)
- imaging studies for Gallstones (e.g.
- abdominal ultrasound)
- assessment for Menorrhagia (e.g.
- complete blood count
- hormonal evaluation).

Case 2: Recurrent Acid Peptic Disease with Nocturnal Burning

Case #2 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: Please tell me

Patient: I had come to you once before—it had improved then. But recently, for the past few days, I'm noticing a slight burning sensation and some pain again.

Doctor: okay

Patient: This time at home—it wasn't really right for me to do this—but I bought those two medicines you had prescribed and took them.

Doctor: hmm

Patient: This time, the gas has reduced quite a bit, but the burning sensation hasn't completely gone away—that's why I've come to see you.

Doctor: its okay

Case 2: Recurrent Acid Peptic Disease with Nocturnal Burning

Case #2 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Gastroesophageal Reflux Disease (GERD) with Recurrent Acid Peptic Disease

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Non-Erosive Reflux Disease (NERD)
- Mild Acute Gastritis

CLINICAL PATHOPHYSIOLOGY & REASONING:

Patient with a history of responsive acid peptic disease presents with breakthrough epigastric and retrosternal burning and pain despite self-resuming prior acid-suppressing medication. While excessive gas symptoms responded to treatment, refractory mucosal burning indicates ongoing gastric hyperacidity or subacute mucosal irritation characteristic of GERD / recurrent gastritis.

RECOMMENDED LAB WORKUP & TESTS:

- Upper GI Endoscopy (Esophagogastroduodenoscopy / EGD)
- Helicobacter pylori Stool Antigen Test or Urea Breath Test
- Complete Blood Count (CBC)
- Trial of optimized Proton Pump Inhibitor (PPI) and mucosal protectant therapy

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Gastroesophageal Reflux Disease (GERD)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Peptic Ulcer Disease (PUD) or Helicobacter pylori infection

CLINICAL REASONING & EVALUATION:

GERD may be causing the recurrent symptoms of burning and pain in the esophagus due to acid reflux

RECOMMENDED LAB WORKUP & TESTS:

- an upper endoscopy
- a 24
- hour pH monitoring study.

Case 3: Classic Gastroesophageal Reflux with Severe Heartburn

Case #3 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: please tell me whats the problem ?

Patient: burning sensation in the stomach

Doctor: Which area ?

Patient: From the chest area.

Doctor: Is it in the upper part of the stomach?

Patient: yes

Doctor: Okay, how many days has it been?

Patient: 2 – 3 year

Doctor: 2 – 3 year? So, have you consulted a doctor for this?

Patient: yes, a doctor had been consulted earlier.

Doctor: Did the doctor do any tests? But this is a report from two years ago, right? Do you have any addictions or habits?

Patient: no

Doctor: Any other illnesses—like sugar, blood pressure, thyroid, or anything else?

Patient: Blood sugar was tested, but nothing significant was there.

Doctor: Does she not take any medicines?

Patient: no.

Patient: If she eats anything else, she feels like vomiting,

Doctor: okay. Any addiction?

Patient: not able to sleep

Doctor: doesn't have any addictions, right?

Patient: no no

Doctor: okay.

Case 3: Classic Gastroesophageal Reflux with Severe Heartburn

Case #3 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Chronic Gastroesophageal Reflux Disease (GERD) with Reflux Dyspepsia

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Functional Dyspepsia (Postprandial Distress Syndrome)
- Secondary Sleep Disturbance / Insomnia due to nocturnal reflux

CLINICAL PATHOPHYSIOLOGY & REASONING:

A 2-3 year chronicity of burning sensation radiating from chest to upper abdomen with postprandial nausea/vomiting feeling strongly indicates chronic gastroesophageal reflux disease (GERD) with reflux-mediated dyspepsia. Nocturnal regurgitation or burning frequently leads to secondary sleep disturbance. Normal fasting blood sugar rules out diabetic gastroparesis.

RECOMMENDED LAB WORKUP & TESTS:

- Upper GI Endoscopy (EGD) to assess for reflux esophagitis or Barrett's esophagus
- 24-hour ambulatory esophageal pH-impedance monitoring
- H. pylori testing (Rapid Urease Test or Stool Antigen)
- Abdominal Ultrasonography (USG)

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Gastroesophageal Reflux Disease (Gerd)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- *None extracted*

CLINICAL REASONING & EVALUATION:

The patient's chief complaint of burning sensation in the stomach and vomiting suggests a possible diagnosis of gastroesophageal reflux disease (GERD). This condition is characterized by the backward flow of stomach acid into the esophagus, causing irritation and inflammation. Other potential causes for these symptoms include gastritis or peptic ulcer disease. However, given the patient's history of burning sensation in the stomach and vomiting, GERD appears to be the most likely diagnosis. To confirm the diagnosis and rule out other conditions, further investigations such as an upper endoscopy or pH monitoring may be recommended. Additionally, lifestyle modifications including dietary changes and elevating the head during sleep can provide symptomatic relief. It is important for the patient to follow up with a healthcare provider for a thorough evaluation and appropriate management of their symptoms.

RECOMMENDED LAB WORKUP & TESTS:

- *None specified*

Case 4: Diet-Induced Dysmotility and Postprandial Distress

Case #4 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: so whats the problem you are having how

Patient: Whatever I eat causes pinching effect in my stomach—like if I eat chickpeas, it causes pinching trouble; even eating peanuts causes discomfort. And I can't tolerate rich food at all—if I eat rich food, even a little meat, it leads to stomach pain

Doctor: Since when has this been happening?

Patient: It has been happening for about two months. Earlier it was mild, but now the irritation in the stomach has increased—it feels like the stomach is constantly burning or churning

Doctor: Bowel movements and urination?

Patient: The bowels don't feel completely clear—even if there is some bowel movement in the morning, it still feels like there's an incomplete, uneasy sensation remaining like dysentery(loose motion with mucus/blood).

Doctor: Any addictions or habits

Patient: no

Doctor: Any other illnesses—like diabetes, blood pressure, or anything else?

Patient: no

Doctor: Are you not taking any medicines for anything?

Patient: No—medicines are ongoing, I mean gyne medicines are ongoing

Doctor: What gynecological problem is there?

Patient: trying to have a baby

Doctor: okay, medicines are given for that, okay Then it's difficult to prescribe too many medicines for you, because if you conceive, these medicines could have side effects on the baby, okay?

Patient: feel like my body has become very weak

Doctor: Yes, that's what I'm saying—some medicines that are usually given to reduce these issues could cause problems for the baby if you conceive, and you might not realize it immediately—you may only find out after 1–2 weeks. So I'm not giving too many medicines; I'll prescribe a small amount for about 3 weeks.

Case 4: Diet-Induced Dysmotility and Postprandial Distress

Case #4 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Diet-Induced Non-Ulcer Dyspepsia with Irritable Bowel Syndrome (IBS-Mixed)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Food Intolerance / Malabsorption (Legume & Lipid Sensitivity)
- Early Pregnancy Medication Caution

CLINICAL PATHOPHYSIOLOGY & REASONING:

Patient presents with a 2-month history of postprandial epigastric pinching, burning, and tenesmus/incomplete evacuation triggered specifically by high-fat foods (meat) and fermentable legumes (chickpeas, peanuts). The combination of upper GI burning and lower GI incomplete evacuation fits IBS-Mixed / dyspepsia overlap. Ongoing conception efforts require non-teratogenic dietary interventions and pregnancy-safe mucosal agents.

RECOMMENDED LAB WORKUP & TESTS:

- Routine Stool Examination & Occult Blood Test
- Stool Routine and Microscopy (for mucus, ova, and cysts)
- Serum Beta-hCG (to confirm pregnancy status prior to medication)
- Complete Blood Count (CBC) and Erythrocyte Sedimentation Rate (ESR)

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Gastric Ulcer

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Peptic Ulcer Disease
- Gastroenteritis
- or Helicobacter pylori infection

CLINICAL REASONING & EVALUATION:

the patient's symptoms are indicative of an ulcerated lesion in the stomach, which can cause discomfort upon ingestion of certain foods and lead to GERD symptoms

RECOMMENDED LAB WORKUP & TESTS:

- endoscopy with biopsy
- stool testing for H. pylori.

Case 5: Chronic Dyspepsia with Suspected Peptic Ulceration

Case #5 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: "What is the problem? Please tell me.

Patient:

Doctor: What problem are you facing now? Please tell me."

Patient: So right now, the problem is that sometimes there is a burning sensation.

Doctor: Which area? In which part of the stomach?

Patient: In the whole abdomen

Doctor: There is a burning sensation throughout the entire abdomen. "How long has this been happening?

Patient: For about 6–7 months, roughly

Doctor: For around 6–7 months?

Patient: yes, Yes, yes—after eating, this whole area feels heavy and tight, like it's stiff or swollen like dhi...

Doctor: Okay—from below the chest?

Patient: Yes, I mean there is pain.

Doctor: I mean, earlier you had a problem and you were advised to get an endoscopy—have you done it?

Patient: No, that hasn't been done

Doctor: it need to be done now

Patient: okay

Doctor: And any new illnesses? Like sugar?

Patient: no no

Doctor: any addiction?

Patient: no no

Doctor: is your appetite normal?

Patient: My appetite isn't very good.

Doctor: is your bowel movement okay?

Patient: that's all right

Doctor: that endroscopy is needed For now, I'm prescribing medicines for 3 weeks. Also, please get an endoscopy done and show me the report once.

Case 5: Chronic Dyspepsia with Suspected Peptic Ulceration

Case #5 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Chronic Dyspepsia with Suspected Peptic Ulcer Disease (PUD)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Erosive Gastritis / Duodenitis
- Functional Postprandial Distress Syndrome

CLINICAL PATHOPHYSIOLOGY & REASONING:

A 6-7 month history of diffuse abdominal burning, early satiety, epigastric tightness, and postprandial pain below the chest indicates chronic peptic disease. The persistence of symptoms and anorexia over half a year warrants urgent diagnostic upper GI endoscopy to evaluate for gastric or duodenal ulceration and rule out mucosal malignancy.

RECOMMENDED LAB WORKUP & TESTS:

- Upper GI Endoscopy (Esophagogastroduodenoscopy) with mucosal biopsy
- H. pylori Rapid Urease Test (RUT)
- Complete Blood Count (CBC) to screen for microcytic anemia
- Abdominal Ultrasonography (USG)

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Gastroesophageal Reflux Disease (GERD)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Peptic Ulcer Disease (PUD)
- Gastritis
- Esophagitis Clinical Reasoning: The patient's symptoms of burning sensation
- pain
- decreased appetite are indicative of GERD

CLINICAL REASONING & EVALUATION:

The patient's symptoms of burning sensation, pain, and decreased appetite are indicative of GERD. These symptoms often occur after eating and can be relieved by taking antacids. However, further investigation is needed to rule out other conditions such as PUD, gastritis, or esophagitis

RECOMMENDED LAB WORKUP & TESTS:

- Endoscopy

Case 6: Choledocholithiasis with Intermittent Biliary Obstruction

Case #6 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: Please tell me what the problem is?

Patient: There is a slight problem in the stomach.

Doctor: What kind of problem are you having? Please tell me. Please tell me what problem you are experiencing

Patient: My appetite is low at times, and my urine appears yellow

Doctor: Okay, since when have these been happening?

Patient: I've been noticing this for about a month.

Doctor: Is it always yellow, or does it become yellow only sometimes?

Patient: No, I had jaundice—since then. If I stay out in the sun or work in the fields during the day, then it happens

Doctor: When you stay out in the sun or work in the fields, your urine becomes yellow—but otherwise, it doesn't happen, right?

Patient: No, after the evening the urine becomes clear again.

Doctor: Then it's not jaundice. Do you have any reports related to the stomach?

Patient: yes there is report

Doctor: any addiction?

Patient: no I don't drink

Doctor: any other addiction?

Patient: bidi

Doctor: any deaseses? thyroid? anything else? Is your appetite normal?

Patient: I don't feel hungry immediately at mealtimes. Sometimes I feel lightheaded.

Doctor: okay. in this report there is joundice but it is 2 month old report. Then again the report should be done at this time, if there is anything, that time there was stone that's why jaundice happened. Then if you're saying that sometimes it is becoming pale/whitish, it could mean the stone has passed. You'll need to do a sonography, and also repeat the jaundice tests. If any problem is found, then the stone may need to be removed. If it has already passed out, then nothing further will be required. Get the tests done and show me the reports

Case 6: Choledocholithiasis with Intermittent Biliary Obstruction

Case #6 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Choledocholithiasis / Biliary Calculi with Intermittent Obstructive Jaundice

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Dehydration-Induced Urine Concentration exacerbating visible chromaturia
- Tobacco Use Disorder (Bidi smoker)

CLINICAL PATHOPHYSIOLOGY & REASONING:

Documented 2-month history of gallbladder/biliary calculus with prior jaundice. Current presentation of episodic dark yellow urine following heavy physical exertion in heat is a combination of transient biliary ductal obstruction (migrating calculus) and heat-induced concentrated dehydration. Repeating hepatobiliary imaging and liver enzymes is critical to confirm stone clearance versus persistent choledocholithiasis.

RECOMMENDED LAB WORKUP & TESTS:

- High-Resolution Abdominal Ultrasonography (USG Hepatobiliary System)
- Liver Function Tests (Total & Direct Bilirubin, AST, ALT, Alkaline Phosphatase)
- Urine Routine and Microscopy (Urobilinogen & Bile Salts)
- MRCP (Magnetic Resonance Cholangiopancreatography) if ductal dilatation is observed

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Choledocholithiasis / Biliary Calculi with Intermittent Obstructive Jaundice

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Acute Cholangitis (Rule out)
- Biliary Colic / Acute Cholecystitis
- Benign Biliary Stricture

CLINICAL REASONING & EVALUATION:

Scleral icterus, yellowish skin discoloration, and dark urine combined with episodic right upper quadrant or epigastric discomfort signify mechanical obstruction of the biliary outflow tract, predominantly caused by migration of gallbladder calculi into the common bile duct (choledocholithiasis). Prompt biliary decompression or extraction prevents suppurative cholangitis.

RECOMMENDED LAB WORKUP & TESTS:

- Magnetic Resonance Cholangiopancreatography (MRCP) — gold standard non-invasive imaging of biliary tree
- Comprehensive Liver Function Tests (LFT) — total/conjugated bilirubin, alkaline phosphatase (ALP), GGT
- High-resolution Abdominal Ultrasonography (USG)
- Endoscopic Retrograde Cholangiopancreatography (ERCP) for therapeutic stone clearance if indicated

Case 7: Chronic Colitis / Suspected Inflammatory Bowel Disease

Case #7 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Patient:

Doctor:

Patient:

Doctor: tell me whats the problem?

Patient: Sir, I've been having stomach problems and for the past 2 years—whenever I get loose motion, it continues for about 15 to 20 days

Doctor: okay Loose motion Is heppening for 2 years? sometime or everytime?

Patient: sometime, in between 1 month it will be perfectly fine then it will start again, for 1 week it might be heavier and then for another week it will be less, and whenever it starts no food gets digested , lower abdomen feels ache, sounds like"kao kao"

Doctor: hmm

Patient: I'm having abdominal cramps, and I'm passing stools frequently

Doctor: Is there any blood with the stool?"

Patient: Sometimes it happens during loose motion

Doctor: so there is blood? This has also been happening for about 2 years, right?

Patient: yes

Doctor: This has been happening for so long—haven't you had any tests or investigations done?"

Patient: About one to one-and-a-half years ago, I got tests done in Berhampore, Murshidabad. I had a USG, and they said I have fatty liver and a gas problem. They prescribed some medicines and also did some blood tests. The blood tests showed there was some infection in the liver and kidneys.

Doctor: any addiction?

Patient: no there is none

Doctor: any other deaseses like sugar ? pressure?

Patient: no there is nothing

Doctor: your apetite is fine?

Patient: apetite is fine.

Doctor: okay

Patient: but the pain in the abdomen is not going away

Case 7: Chronic Colitis / Suspected Inflammatory Bowel Disease

Case #7 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Inflammatory Bowel Disease (Suspected Ulcerative Colitis) vs Chronic Infectious Colitis

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Non-Alcoholic Fatty Liver Disease (NAFLD, incidental on USG)
- Secondary Colonic Malabsorption / Dysbiosis

CLINICAL PATHOPHYSIOLOGY & REASONING:

A 2-year history of episodic bloody diarrhea lasting 2-3 weeks at a time with lower abdominal cramping, borborygmi, and tenesmus is a red-flag presentation strongly suggestive of inflammatory bowel disease (such as Ulcerative Colitis) or relapsing infectious colitis (e.g., amebiasis or post-dysenteric colitis). Colonoscopy with histopathology is mandatory to evaluate the extent of mucosal ulceration.

RECOMMENDED LAB WORKUP & TESTS:

- Full Colonoscopy with Ileo-Colonic Mucosal Biopsies
- Fecal Calprotectin (biomarker of intestinal inflammation)
- Stool Routine, Occult Blood, Microscopy & Stool Culture
- Complete Blood Count (CBC), ESR, and C-Reactive Protein (CRP)

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Irritable Bowel Syndrome (IBS)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Gastroenteritis for the loose motion
- abdominal cramps
- Hepatitis for the liver infection

CLINICAL REASONING & EVALUATION:

The patient's chief complaint includes stomach problems with loose motion lasting for 15 to 20 days, abdominal cramps, frequent stools, pain in the lower abdomen, and a history of fatty liver and gas problem. The blood tests revealed an infection in the liver and kidneys. Based on these findings, the primary diagnosis is likely Irritable Bowel Syndrome (IBS). IBS is characterized by recurrent abdominal pain or discomfort associated with altered bowel habits. Secondary diagnoses could include Gastroenteritis for the loose motion and abdominal cramps, and Hepatitis for the liver infection. However, further investigation is needed to confirm these diagnoses. The recommended tests are: 1. Colonoscopy to evaluate the colon for any abnormalities or signs of IBS. 2. Liver function tests (LFTs) to assess the liver's health and rule out other causes of liver inflammation. 3. Stool culture to check for bacterial infections that could be causing the gastrointestinal symptoms. It is important to conduct these tests to confirm the diagnoses and develop an appropriate treatment plan for the patient.

RECOMMENDED LAB WORKUP & TESTS:

- Colonoscopy to evaluate the colon for any abnormalities or signs of IBS.
- Liver function tests (LFTs) to assess the liver's health
- rule out other causes of liver inflammation.

- Stool culture to check for bacterial infections that could be causing the gastrointestinal symptoms

Case 8: Refractory Epigastric Pain with Acid Peptic Dyspepsia

Case #8 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: what is the prblem tell me

Patient: Not able to eat properly—whatever is eaten turns into gas.

Doctor: hmm

Patient: A few days ago, blood level had dropped and needed a blood transfusion.

Doctor: When did this happen? When did your blood level drop?

Patient: My blood level dropped about 2 weeks ago, and I was given blood 2–4 days ago.

Doctor: How long have you been unable to eat properly?

Patient: For about 2 weeks—since the time my blood level dropped

Doctor: Before this, did you have any known illnesses—like diabetes, blood pressure, thyroid, or anything else?

Patient: yes there is pressure

Doctor: is medication taken for that?

Patient: yes sir I had consulted a doctor earlier, and I had started taking medicines.

Doctor: for what? Pressure? for gas and I was not able to sleep.

Patient: So sir, my mother stays in our village. After going there, she was fine for a few days, but then complications started again. So we went to a doctor at Apollo Hospitals and consulted here. The doctor said it's a stomach-related problem, but coming here frequently might be difficult.

Doctor: any addiction?

Patient: no

Doctor: how is the apetite?

Patient: She's not able to eat properly.

Doctor: is it for the past 2 week or before that?

Patient: before that

Doctor: is there any weight loss? It's visibly noticeable—her blouse has become loose

Patient: no

Doctor: Is there any vomiting or anything like that?

Patient: no

Doctor: bowel movement is fine?

Patient: The stool is somewhat blackish in color.

Doctor: blackish?

Patient: after the blood transfution

Doctor: after the blood transfution it became blackish? Is she taking any iron supplements or medicines?

Patient: yes, we are only feeding her that , that is written here

Doctor: Oh, that's because of taking iron supplements—the stool can become black due to that. Alright, please lie down—I'll examine your abdomen.

Patient: my body is becoming very weak

Doctor: Only one time of blood was given? And after the transfusion, you haven't done another blood test, right?

Patient: no there was blood test in the hospital and the doctor there told that everything is fine.

Doctor: that report was not given?

Patient: no they have written that its fine

Doctor: okay

Case 8: Refractory Epigastric Pain with Acid Peptic Dyspepsia

Case #8 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Severe Anemia Secondary to Suspected Upper Gastrointestinal Bleed

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Iron Supplementation-Associated Dyspepsia & Melenoid Stool
- Hypertension
- Rule out Gastric / Colonic Malignancy or Bleeding Peptic Ulcer

CLINICAL PATHOPHYSIOLOGY & REASONING:

An elderly patient with acute severe anemia necessitating packed red blood cell transfusion combined with anorexia, early satiety, and blackish stool represents a medical emergency. While iron therapy produces black stools, the sudden severe blood loss in an elderly patient mandates comprehensive endoscopic evaluation to identify and exclude bleeding peptic ulcers, vascular ectasias, or occult GI neoplasms.

RECOMMENDED LAB WORKUP & TESTS:

- Upper GI Endoscopy (EGD) and Colonoscopy
- Repeat Complete Blood Count (CBC) with Peripheral Blood Smear
- Serum Ferritin, Iron, and Total Iron Binding Capacity (TIBC)
- Fecal Occult Blood Test (iFOBT)

⚙ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Malabsorption

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Iron deficiency Anemia due to chronic blood loss or malabsorption

CLINICAL REASONING & EVALUATION:

the patient's gastrointestinal symptoms are indicative of malabsorption, which can lead to vitamin and mineral deficiencies including iron, resulting in anemia. Confirmatory investigations would include serologic testing for celiac disease antibodies such as anti-tissue transglutaminase (anti-TTG) or endomysial antibody (EMA), and further evaluation of iron status through complete blood count (CBC) with peripheral smear examination and serum iron studies.

RECOMMENDED LAB WORKUP & TESTS:

- *None specified*

Case 9: Acute Upper/Lower Gastrointestinal Hemorrhage

Case #9 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: please tell me whats the problem?

Patient: sir for the last 7 months there is pain in my abdomen like burning sensation

Doctor: where?

Patient: on the top side

Doctor: okay

Patient: I went up for checking to doctor , did the endroscopy also 4 months ago , I had an ulcer in my stomach—I tested positive for it

Doctor: okay

Patient: I have taken all the medicines and completed the full course.

Doctor: hmm

Patient: But still, I notice blood passing with the stool.

Doctor: Since when have you been passing blood in your stool?

Patient: since yesterday The stool is completely black like tar, and there is blood mixed with it, First, in the morning when I passed stool, there was fresh blood. After that, it has been completely black.

Doctor: then getting admiited will be better. Endoscopy is also needed, blood transfusion might also be needed, okay It's quite possible that the bleeding is coming from here and hasn't stopped yet. Okay, we'll need to check that. Are you on an empty stomach, or have you eaten?

Patient: I have eaten just now

Doctor: then it's a problem, I'm writing an admission note for you.

Patient: Sir, I'm saying—on both sides it feels like something is pricking, like a needle, and there is a burning sensation.

Doctor: okay This report is very old , 4 month old

Patient: yes sir Sir I have a kid also

Doctor: just get admitted for 1 or 2 days , hopefully it will be done by tomorrow, if your endoscopy is needed then also it cants be done before the evening, when did u ate?

Patient: 11

Doctor: so 4 hours are needed atleast , so endoscopy will be done in the evening and if blood transfusion is needed then till tomorrow morning it will be needed, we will look after that later after looking at the report

Patient: okay

Doctor: atleast getting admitted for 1 day is needed when there is black stool perticularly

Patient: sir medicine?

Doctor: but medicine could not controlled the problem in your case

Case 9: Acute Upper/Lower Gastrointestinal Hemorrhage

Case #9 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Acute Upper Gastrointestinal Hemorrhage Secondary to Bleeding Gastric / Peptic Ulcer

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Severe Acute Blood Loss Anemia
- Refractory Peptic Ulcer Disease (Rule out H. pylori resistance / NSAID use)

CLINICAL PATHOPHYSIOLOGY & REASONING:

Patient with endoscopic confirmation of gastric ulcer 4 months ago now presents with classic melena (black tarry stool) and hematochezia, signaling brisk acute upper gastrointestinal hemorrhage. This is a life-threatening complication requiring emergency inpatient stabilization, hemodynamic monitoring, intravenous proton pump inhibitors, and urgent therapeutic endoscopy.

RECOMMENDED LAB WORKUP & TESTS:

- Emergency Diagnostic & Therapeutic Upper GI Endoscopy (EGD)
- Serial Complete Blood Count (CBC) & Hematocrit Monitoring
- Blood Grouping, Typing, and Crossmatch for immediate PRBC transfusion
- Coagulation Profile (PT/INR, aPTT) and Serum Electrolytes

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Melena (Upper GI Bleed)

SECONDARY / DIFFERENTIAL DIAGNOSES:

- peptic ulcer disease with complications such as bleeding or perforation
- ischemic colitis
- diverticulosis with bleeding
- or malignancy in the colon or stomach. Further evaluation is needed through endoscopy
- possibly a blood transfusion to address the acute presentation of gastrointestinal bleeding.

CLINICAL REASONING & EVALUATION:

The patient's chief complaint includes abdominal pain with a burning sensation and blood in the stool, which is consistent with gastrointestinal bleeding. The history of present illness reveals an ulcer in the stomach confirmed by endoscopy 4 months ago. The recent change in stool color to black indicates melena, suggesting active upper gastrointestinal bleeding. The location of the pain on both sides suggests possible involvement of the colon or small intestine. A differential diagnosis for this patient may include peptic ulcer disease with complications such as bleeding or perforation, ischemic colitis, diverticulosis with bleeding, or malignancy in the colon or stomach. Further evaluation is needed through endoscopy and possibly a blood transfusion to address the acute presentation of gastrointestinal bleeding.

RECOMMENDED LAB WORKUP & TESTS:

- *None specified*

Case 10: Congestive Heart Failure with Secondary Visceral Congestion

Case #10 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: tell me whats the problem?

Patient: There is abdominal pain, and when food stays here, it feels swollen or bloated.

Doctor: Does your abdomen feel swollen or bloated?

Patient: yes My abdomen feels swollen, and my body—hands and feet—also get swollen

Doctor: Since when has this been happening?

Patient: near about 1 year

Doctor: 1 year?

Patient: hmm

Doctor: any other deaseses like sugar , pressure, thyroid?

Patient: yes there is thyroid

Doctor: medication taken for thyroid?

Patient: yes 100 power

Doctor: okay, any addiction?

Patient: no there is none

Doctor: bowel movement?

Patient: I have bowel movements once every two days.

Doctor: that problem is there for way long right?

Patient: bowel movement is not getting cleared for long, for like 3 to 4 moths its happening like this

Doctor: 3 – 4 moths , right? hmm, u told no addiction is there right? Is your apetite working fine?

Patient: no there is probelm

Doctor: any test was there? for this stomach problem?

Patient: yes there was but I didn't bring the report

Doctor: if you didn't bring it then what can be done, As u didn't bring it u have to do it , there is no otherwise

Patient: okay , And I'm not able to digest food properly; my stomach stays bloated

Doctor: The upper part of the abdomen, right? I'll examine and check it.

Case 10: Congestive Heart Failure with Secondary Visceral Congestion

Case #10 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Hypothyroidism-Induced Gastrointestinal Dysmotility with Peripheral Myxedema

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Chronic Functional Dyspepsia with Abdominal Distension
- Non-Pitting Peripheral Edema secondary to Thyroid Dysfunction

CLINICAL PATHOPHYSIOLOGY & REASONING:

Patient with established hypothyroidism on 100 mcg levothyroxine presents with a 1-year history of persistent abdominal bloating, generalized swelling of hands and feet, and chronic constipation. Suboptimally controlled hypothyroidism causes intestinal hypomotility, delayed gastric emptying, and mucopolysaccharide deposition in dermis and connective tissue (myxedema). Re-evaluating thyroid dosage and metabolic status is essential.

RECOMMENDED LAB WORKUP & TESTS:

- Comprehensive Thyroid Panel (Free T3, Free T4, Ultrasensitive TSH)
- Abdominal Ultrasonography (USG)
- Serum Electrolytes, Creatinine, and Liver Function Tests
- Complete Blood Count (CBC)

⚙️ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Celiac disease

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Irritable Bowel Syndrome (IBS) or Inflammatory Bowel Disease (IBD)

CLINICAL REASONING & EVALUATION:

The patient's chief complaint includes abdominal pain and bloating, as well as swelling of the abdomen and extremities. The patient also reports infrequent bowel movements and difficulty digesting food properly. Based on these symptoms, a possible primary diagnosis is Celiac disease. Secondary diagnoses could include Irritable Bowel Syndrome (IBS) or Inflammatory Bowel Disease (IBD). To confirm the diagnosis, recommended tests include blood tests for antibodies associated with celiac disease, such as anti-endomysial antibody test and anti-tissue transglutaminase antibody test. Additionally, an endoscopy may be performed to evaluate the small intestine for damage characteristic of celiac disease.

RECOMMENDED LAB WORKUP & TESTS:

- blood tests for antibodies associated with celiac disease
- such as anti
- endomysial antibody test
- anti
- tissue transglutaminase antibody test. Additionally
- an endoscopy may be performed to evaluate the small intestine for damage characteristic of celiac disease.

Case 11: Thoracoabdominal Radiculopathy vs Acute/Subacute Pancreatitis

Case #11 of 11 • Whole Consultation Transcript

WHOLE CONSULTATION TRANSCRIPT

Doctor: tell me whats the problem ?

Patient: Doctor, whatever I eat, it feels like...

Doctor: What does it feel like?

Patient: It feels heavy on this side, near the side of the abdomen (towards the flank).

Doctor: Since when has this been happening?

Patient: For many days.

Doctor: it doesn't seem like stomach problem it looks like nerve problem , okay? But is there any test doen for stomach?

Patient: yes, 1 time I have done the report

Doctor: what report?

Patient: I did the ultrasonography said it was fat

Doctor: That usually happens if there is abdominal fat, but you don't have that. but there was no endoscopy?

Patient: no there was not

Doctor: any other deaseses , sugar , pressure, anything?

Patient: nothing

Doctor: any addiction?

Patient: no

Doctor: Is everything else normal—your appetite, bowel movements, and urination?

Patient: yes But I feel very little hunger—I hardly feel hungry at all.

Doctor: okay, I will do the test and all once

Patient: cannot eat anything also

Doctor: okay, I don't think it is a stomach problem specific, I think it is a problem related to nerve , later you need to get checked up by a nerve doctor , 1st the report should be done here then

Patient: okay

Doctor: I am giving you medicine for like 3 week , okay?

Case 11: Thoracoabdominal Radiculopathy vs Acute/Subacute Pancreatitis

Case #11 of 11 • Clinical Diagnostic Evaluation

📋 Ground Truth - Claude

PRIMARY DIAGNOSIS:

Thoracoabdominal Radiculopathy / Intercostal Nerve Entrapment Syndrome vs Functional Flank Dyspepsia

SECONDARY / DIFFERENTIAL DIAGNOSES:

- Abdominal Wall Pain Syndrome (Carnett's Syndrome)
- Mild Functional Anorexia

CLINICAL PATHOPHYSIOLOGY & REASONING:

Patient reports localized heaviness and discomfort in the abdominal flank rather than epigastric burning, without acute peritoneal signs. Negative intra-abdominal ultrasound findings and lack of gastrointestinal alarm symptoms support the clinician's diagnostic impression of a neuropathic or musculoskeletal etiology (intercostal neuralgia or thoracoabdominal radiculopathy). Neurological assessment and confirmatory imaging are indicated.

RECOMMENDED LAB WORKUP & TESTS:

- Focused Thoracolumbar Spine & Abdominal Wall Physical Examination (Carnett's test)
- Repeat High-Resolution Abdominal & Retroperitoneal Ultrasonography
- Spinal MRI / Nerve Conduction Study (if radicular symptoms persist)
- Basic Metabolic Panel and Routine Urinalysis (to exclude renal colic)

⚙ OpenBioLLM-8B Inference

PRIMARY DIAGNOSIS:

Acute Pancreatitis

SECONDARY / DIFFERENTIAL DIAGNOSES:

- *None extracted*

CLINICAL REASONING & EVALUATION:

The patient's chief complaint of feeling heavy on one side near the flank and decreased appetite is suggestive of acute pancreatitis. This condition can cause epigastric pain radiating to the back, which may present as heaviness in the affected quadrant. The loss of appetite is a common symptom associated with inflammation of the pancreas. However, further diagnostic tests are necessary to confirm the diagnosis. Recommended Tests: - Serum amylase and lipase levels - CT scan of the abdomen

RECOMMENDED LAB WORKUP & TESTS:

- Serum amylase
- lipase levels
- CT scan of the abdomen